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22F-3277

BCS-3C

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LAB 5

(Total Marks 15)

Task 1: Draw and run following code and write description of execution.

```
org 0x100
jmp start

multiplicand: db 13
multiplier: db 5
result: db 0

start: mov cl, 4
      mov bl, [multiplicand]
      mov dl, [multiplier]

checkbit: shr dl, 1
         jnc skip
         add [result], bl

skip: shl bl, 1
     dec cl
     jnz checkbit

mov ax, 0x4c00
int 0x21
```

Also fill the following table after execution of each loop in binary form.

Count	Multiplicand	Multiplier	Result
4	11010	0010	1101
3	110100	0001	1101
2	1101000	0000	01000001
1	11010000	0000	01000001

We take the first digit of the multiplier and multiply it with the multiplicand. If the digit of multiplier is one the answer will be multiplicand itself. So, it is placed in result. Before multiplying with the next digit, as we do in common multiplication a cross is placed at the right most place on next line and the result is shifted to left, so in this we now must shift multiplicand to the left so we will use left shift. Now we again check the multiplier bit and if it is zero perform no addition. Left shift the multiplicand before the next bit of multiplier is tested. And same as this then we add and place it in result and for next we will again shift it to left as in simple multiplication we place two crosses so here it will be second shift so this process is repeated and until final result.

Task 2: Write assembly language code to multiply two-word size numbers:

- **Multiply 600* 350**
- **Use shift right, shift left and rotate carry left operations**

```
org 0x0100
```

```
mov cl, 16
```

```
mov dx, [multiplier]
```

```
checkbit:
```

```
shr dx, 1
```

```
jnc skip
```

```
mov ax, [multiplicand]
```

```
add [result], ax
```

```
mov ax, [multiplicand+2]
```

```
adc [result+2], ax
```

```
skip:
```

```
shl word [multiplicand], 1
```

```
rcl word [multiplicand+2], 1
```

```
dec cl
```

```
jnz checkbit
```

```
mov ax, 0x4c00
```

```
int 0x21
```

```
multiplicand: dd 600
```

```
multiplier: dd 350
```

```
result: dd 0
```

AX	4C00	SI	0000	CS	19F5	IP	0127	Stack	+0 0000	Flags	7244
BX	0000	DI	0000	DS	19F5				+2 20CD		
CX	0000	BP	0000	ES	19F5	HS	19F5		+4 9FFF	OF DF IF SF ZF AF PF CF	
DX	0000	SP	FFFE	SS	19F5	FS	19F5		+6 EA00	0 0 1 0 1 0 1 0	

S or SI or SYM										1									
CMD >S										0 1 2 3 4 5 6 7									
0124 B8004C MOV AX,4C00										DS:0000 CD 20 FF 9F 00 EA F0 FE									
0127 CD21 INT 21										DS:0008 AD DE 1B 05 C5 06 00 00									
0129 0000 ADD [BX+SI],AL										DS:0010 18 01 10 01 18 01 92 01									
012B 58 POP AX										DS:0018 01 01 01 00 02 FF FF FF									
012C 025E01 ADD BL,[BP+01]										DS:0020 FF FF FF FF FF FF FF FF									
012F 0000 ADD [BX+SI],AL										DS:0028 FF FF FF FF EB 19 C0 11									
0131 50 PUSH AX										DS:0030 A2 01 14 00 18 00 F5 19									
0132 3403 XOR AL,03										DS:0038 FF FF FF FF 00 00 00 00									
0134 00D2 ADD DL,DL										DS:0040 05 00 00 00 00 00 00 00									
										DS:0048 00 00 00 00 00 00 00 00									

2	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	0	P4..u.à Lt. F..			
DS:0131	50	34	03	00	D2	75	04	85	C0	74	1C	C7	46	DC	00	00	A^"â}.t .iF>H:F÷			
DS:0141	8E	5E	FC	83	7D	0E	00	74	09	8B	46	F2	48	3B	46	F6	~.q..δ.θ B.1LεFF'i			
DS:0151	7E	08	B8	01	00	EB	05	E9	42	01	31	C0	89	46	E2	8B	F÷TαTα ^ ‡.†i7iG.			
DS:0161	46	F6	D1	E0	D1	E0	C5	5E	D8	01	C3	8B	37	8B	47	02	èF■àLu.à ÷t.A†âi.			
DS:0171	89	46	FE	85	C0	75	04	85	F6	74	12	8E	D8	83	7C	0C				

Task 3: Write a program to count the number of 1's in a word size four variable in single label.

- **num: dw 65535, 65535, 39321, 39321**
- **Use Rotate Left (ROL) Operation**
- **Count the numbers of 1's and save it into another variable with its address.**

```
org 0x0100
```

```
mov ch,4
mov si, num
mov bx,0
```

```
outer:
    mov ax, [si]
    mov cl, 16
```

```
checkone:
    rol ax, 1
    jnc skip
    add bx, 1
```

```
skip:
    dec cl
    jnz checkone

    add si, 2
    dec ch
    jnz outer
```

```
mov [onescount], bx
```

```
mov ax,0x4c00
int 0x21
```

```
num: dw 65535,65535,39321,39321
onescount: dw 0
```

DOSBox 0.74, Cpu speed: 3000 cycles, Frameskip 0, Program:...																-		□		✕	
AX	0000	SI	0000	CS	19F5	IP	0100	Stack	+0 0000	Flags	7202										
BX	0000	DI	0000	DS	19F5				+2 20CD												
CX	0000	BP	0000	ES	19F5	HS	19F5		+4 9FFF	OF	DF	IF	SF	ZF	AF	PF	CF				
DX	0000	SP	FFFE	SS	19F5	FS	19F5		+6 EA00	0	0	1	0	0	0	0	0				
S or SI or SYM																					
CMD >S																					
								DS:0131	30	00	02	85	D2	75	04	85					
								DS:0139	C0	74	1C	C7	46	DC	00	00					
0100	B504			MOV	CH,04				DS:0141	8E	5E	FC	83	7D	0E	00	74				
0102	BE2901			MOV	SI,0129				DS:0149	09	8B	46	F2	48	3B	46	F6				
0105	BB0000			MOV	BX,0000				DS:0151	7E	08	B8	01	00	EB	05	E9				
0108	8B04			MOV	AX,[SI]				DS:0159	42	01	31	C0	89	46	E2	8B				
010A	B110			MOV	CL,10				DS:0161	46	F6	D1	E0	D1	E0	C5	5E				
010C	D1C0			ROL	AX,1				DS:0169	D8	01	C3	8B	37	8B	47	02				
010E	7304			JNC	0114				DS:0171	89	46	FE	85	C0	75	04	85				
0110	81C30100			ADD	BX,0001				DS:0179	F6	74	12	8E	D8	83	7C	0C				
2	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	0					
DS:0131	30	00	02	85	D2	75	04	85	C0	74	1C	C7	46	DC	00	00	0. .àµu.à Lt. F■. .				
DS:0141	8E	5E	FC	83	7D	0E	00	74	09	8B	46	F2	48	3B	46	F6	~^"â}. .t .iF2H:F÷				
DS:0151	7E	08	B8	01	00	EB	05	E9	42	01	31	C0	89	46	E2	8B	~. .δ.0 B.1LeFFi				
DS:0161	46	F6	D1	E0	D1	E0	C5	5E	D8	01	C3	8B	37	8B	47	02	F÷µµµµµµ µ. i7IG.				
DS:0171	89	46	FE	85	C0	75	04	85	F6	74	12	8E	D8	83	7C	0C	eFµàµ.à ÷t.À+â!i.				